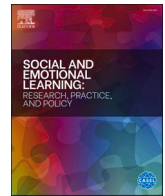




Contents lists available at ScienceDirect

Social and Emotional Learning: Research, Practice, and Policy

journal homepage: www.journals.elsevier.com/social-and-emotional-learning-research-practice-and-policy



Empirical/Meta-analyses

A systematic review and meta-analysis of the effects of universal school-based SEL programs in the United States: Considerations for marginalized students[☆]

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ARTICLE INFO

Keywords:

Social and Emotional Learning (SEL)
Marginalized students
Meta-analysis
Education in the United States

ABSTRACT

The present study systematically reviewed and meta-analyzed the evidence for universal school-based SEL programming within the United States to understand nation-specific trends in SEL implementation and effectiveness, and provide specific recommendations for marginalized students- particularly those with minority gender-, racial- ethnic, linguistic-, and disability identities, within SEL. Analyzing the effects of 90 universal school-based social and emotional learning studies, reflecting 47 discrete SEL programs, and 20,626 students in grades K-12 in the United States, from January 1, 2008 through December 31, 2020, we report that students who participate in universal school-based SEL programs experienced significantly improved academic achievement, school functioning, social and emotional skills, attitudes, behaviors, and perceptions of school climate and safety. Furthermore, programs in the United States benefited students in elementary and secondary school equally, and had the strongest outcomes for students when teachers were the program implementors. SEL programs in the United States were varied in their program features, duration, and dosage, had limited engagement with families and community, and did not show evidence of an overall follow-up effect 6 months or longer after a program ends. SEL programs were found to be equally efficacious for boys and girls meta-analytically, and systematic review of available studies revealed preliminary evidence of differential and positive effects for students with marginalized racial and ethnic identities. Limited representation of marginalized linguistic, sexual, and ability identities inhibited exploration of additional effects analytically, and descriptive results, alongside opportunities for increasing reporting and analyses are discussed.

Impact Statement: This study's findings provide robust evidence for the beneficial effects of universal school-based SEL programs for K-12 students in the United States. Systematic review and meta-analysis of 90 studies, reflecting 47 discrete SEL programs, and more than 20,000 K-12 students, found that participation in an SEL program significantly improved student academic achievement, school functioning, social and emotional skills, attitudes, behaviors, and perceptions of school climate and safety.

At the time we prepared this manuscript, diminished social and emotional health among youth had become such a widespread issue in

the United States that the American Academy of Pediatrics (AAP), the American Academy of Child and Adolescent Psychiatry (AACAP), and

[☆] This research was funded by grants to Dr. Cipriano from the Oak Foundation (OCAY-19-407 and OFIL-22-040)

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<https://doi.org/10.1016/j.sel.2024.100029>

Received 29 August 2023; Received in revised form 24 January 2024; Accepted 14 February 2024

Available online 17 February 2024

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the Children's Hospital Association (CHA) declared a "National State of Emergency in Children's Mental Health" which recognized that both the COVID-19 pandemic and the ongoing social injustices have impacted nation's young people (American Academy of Pediatrics, 2021). With nearly three-quarters of public schools having reported an increase in the number of students requesting mental health support (Irwin et al., 2023), and even though the American Rescue Act Plan (ARP) allocated \$128 billion toward "staffing decisions that will help address students' social, emotional, mental health, and academic needs" (The White House, 2022), at the currently, only a single state, Idaho, and Washington, D.C. met the recommended ratio of one school psychologist for every 500 students (Hopeful Futures Campaign, 2022).

Universal school-based social and emotional learning programs are one cost effective and widely available approach for helping students develop the necessary skills to navigate the mental health crisis within and outside of schools (Belfield et al., 2015; Greenberg, 2023). Universal School-Based (USB) social and emotional learning (SEL) programs support whole child development, emphasizing that comprehensive student development requires nurturing student social and emotional skill development concurrently with academic instruction. USB SEL programs are programs that are delivered school-wide or to a whole classroom to support the development of student intra- and inter- personal skills. USB SEL programs promote the physical and psychological health for all students rather than targeting the social and emotional needs of a specific group or population of students (i.e., boys only, special education students; Cipriano, Naples, et al., 2023; Cipriano, Strambler, et al., 2023). Universal school-based SEL interventions have a legacy of demonstrated success in improving students' skills and outcomes in the short and long term (Durlak et al., 2011; Durlak et al., 2022; Greenberg, 2023; Taylor et al., 2017; Wigelsworth, 2020). Most recently, our registered review of over 400 experimental studies, across 53 countries, reported that students who participated in USB SEL interventions demonstrated significant improvements in school climate and safety, civic attitudes, and behaviors, SEL skills, peer relationships, attitudes and beliefs, prosocial behaviors, school functioning, and academic achievement, compared to students not participating in these interventions (Cipriano, Strambler, et al., 2023). Findings also illustrated that students demonstrated significant reductions in emotional distress and externalizing behaviors after participating in SEL. The evidence from our international review clearly suggests that SEL programs have the potential to support students nationwide with the skills and strategies necessary to promote mental health and well-being in school (Cipriano, Strambler, et al., 2023).

Marginalized students and SEL

While the mental health crisis in the United States is not bound to a particular region, or demographic, the data available suggests it has not impacted everyone equally. The CDC for example, has highlighted the variability in social-emotional health in the U.S. which widely differs among age, race, class, gender, ethnicity, sexuality, disability, and geographical location with some groups being more susceptible to suicidal thoughts (CDC, 2022). In 2021, risk of attempting suicide was greater among Black students, female students, and students who identify as Lesbian, Gay, Bisexual, Questioning, Other Non-Heterosexual Identity (LGBQ+) (CDC, 2023). Additionally, prolonged feelings of "sadness or hopelessness" was highest among Hispanic and multiracial students when compared to students of other races and ethnicities (CDC, 2023, p. 60). These same adverse feelings were shared among nearly 70% of LGBQ+ students in 2021, with more than a third of LGBQ+ students planning to commit suicide and nearly a quarter of these students attempting suicide (CDC, 2023).

Concurrent to these devastating mental health statistics, U.S. K-12 education has been pushed to the frontline of partisan discourse, with some schools falling victim to policies of exclusion (Ridings, 2020; Stanford, 2022; Woo et al., 2023). Many states across the nation, for

example, have passed legislation limiting the presence of gender identity and sexual orientation in classroom discourse (see Rosky, 2017; Ridings, 2020). There has also been a resurgence of attacks against African American history, with particular attention on Advanced Placement courses (Najarro, 2023). Consequently, RAND Corporation found that teachers were concerned that limitations placed on how students learn about race and gender could impact students' academic development (i.e., "critical thinking skills") and may also jeopardize their social-emotional health as well as the development of their social and emotional skill set (i.e., "empathy"; Woo et al., 2023, p. 18). These ongoing attacks on student identities create hostile learning environments for students with minoritized identities.

Importantly, despite the success of SEL programs on the social and emotional development of K-12 students in general, numerous reports, including our own, have documented limited representation of discrete populations of students, particularly those who are at greatest risk for diminished mental health, in experimental and quasi experimental evaluations of SEL (Cipriano, Naples et al., 2023; Cipriano & McCarthy, 2023; Daley & McCarthy, 2021; Rowe & Trickett, 2018). Although the reason for this under representation can be linked to sample sizes needs for statistical power, SEL is situated within the systems of power and privilege of the education system, and students' intersectional, overlapping, and interconnected identities (Artiles et al., 2016; Crenshaw, 1991) influence their experiences of SEL programming in context (Jagers et al., 2019; Rappolt-Schlichtmann et al., 2024; Rivas-Drake et al., 2020). Despite comprehensive evaluations of SEL increasing (Durlak et al., 2022; Greenberg, 2023), the meta-analytic impact that SEL programs have on the social and emotional functioning of marginalized students in universal classroom environments is largely unknown. Experimental designs in intervention research often do not report how outcomes differ for subgroups of students based on marginalized gender, race/ethnicity, native language learner status, socioeconomic status, LGBTQIA+ , and/or disability status, in part due to statistical power needs (Cipriano, Naples et al., 2023; Cipriano, Strambler et al., 2023; Daley & McCarthy, 2021). For example, in a 2022 systematic review of universal SEL interventions in middle and high school, Daley and McCarthy found that only 11.4% of studies considered students with disabilities, and out of those studies, only five conducted subgroup analyses to examine disparities in student experience. Likewise, an additional review by Cipriano, Naples, and colleagues (2023) appraised the experiences of elementary students within 269 studies of universal SEL interventions and identified that only 27.5% of studies conducted analyses using race/ethnicity as a moderator or covariate with a majority using broad racial and ethnic identifiers (i.e., Black, white, non-white). Findings from an international meta-analysis of universal SEL interventions in schools confirm a systemic lack of reporting and consideration of student identity variables and a dearth of evidence on the experiences of historically marginalized populations within USB SEL (Cipriano, Strambler et al., 2023). Understanding the interaction of student identities with SEL in the context of school systems and the power and privileges steeped therewith is a necessary, and complex, area of inquiry for the field of SEL to address (Cipriano & McCarthy, 2023).

Towards this end, the present study aimed to systematically review and meta-analyze the evidence for universal school-based SEL programming within the United States to understand nation-specific trends in SEL implementation and effectiveness, and provide specific recommendations for marginalized students within SEL. The primary research aim was to examine if and how USB SEL programs for students in grades K-12 enhance student social, emotional, and academic outcomes. We expected this review to yield significant mean effects across all SEL competencies and outcomes (Hypothesis 1; Cipriano, Naples et al., 2023; Durlak et al., 2011), and as follow-up effect data are available, we hypothesize significant mean effects on all available long-term outcomes (Hypothesis 2: Cipriano, Strambler, et al., 2023; Taylor et al., 2017). In addition, regarding features of SEL programs, we hypothesized that the

effects of students' SEL competencies and outcomes would be strongest for programs implemented by the classroom teacher (Hypothesis 4: Cipriano, Strambler, et al., 2023; Durlak et al., 2011). Further, we hypothesized there would be no difference in the effects of SEL programs at the elementary or secondary school levels (Hypothesis 5: Cipriano, Strambler, et al., 2023). Finally, these analyses set the stage for the detailed examination of the effect of SEL programs in the US for students with marginalized identities. We hypothesized that the effect of students' SEL competencies and outcomes would vary across subgroups based on students' identity (Hypothesis 6: Cipriano & McCarthy, 2023; Daley & McCarthy, 2021).

Methods

This meta-analysis is a secondary data analysis of a subset of studies exclusively conducted in the United States from Cipriano, Strambler, and colleagues' 2023 registered international report of experimental and quasi-experimental USB SEL interventions from 2008 through 2020. In Cipriano, Strambler, et al., 2023, the sample reflected 53 countries, including 424 studies, and 252 discrete SEL interventions; of this sample, 258 studies from 45 countries reported statistical results for the available effect sizes for the meta-analysis. In the current paper, we draw results from the 90 studies conducted in the United States, and the four studies that provided follow-up effects after the six months from the SEL program ending in the United States. This meta-analysis adhered to

best-practices by following rigorous PRISMA guidelines (Page et al., 2021). Furthermore, the dataset used in this study has been made publicly available to further the goals of open science <<https://osf.io/6pqx2>> . Additional details and the pre-registered codebook are available at <<https://osf.io/cszn8>> .

Inclusion criteria and exclusion criteria

Inclusion criteria

This review included published and unpublished reports of universal interventions that teach at least one intrapersonal and interpersonal skill (e.g., self-management and communication skills). We included studies that were (a) randomized or quasi-experimental, (b) used a control group (wait list, or attention placebo), (c) involved at least six sessions if the intervention was classroom-based or at least four months if a whole school approach was taken; (d) took place in the United States between January 1, 2008 and December 31, 2020 (including online first publications), and (e) provided data for the calculation of effect sizes either in print or provided by authors.

Exclusion criteria

Studies without a control condition, when SEL was not the primary component of the intervention, and studies that targeted specific

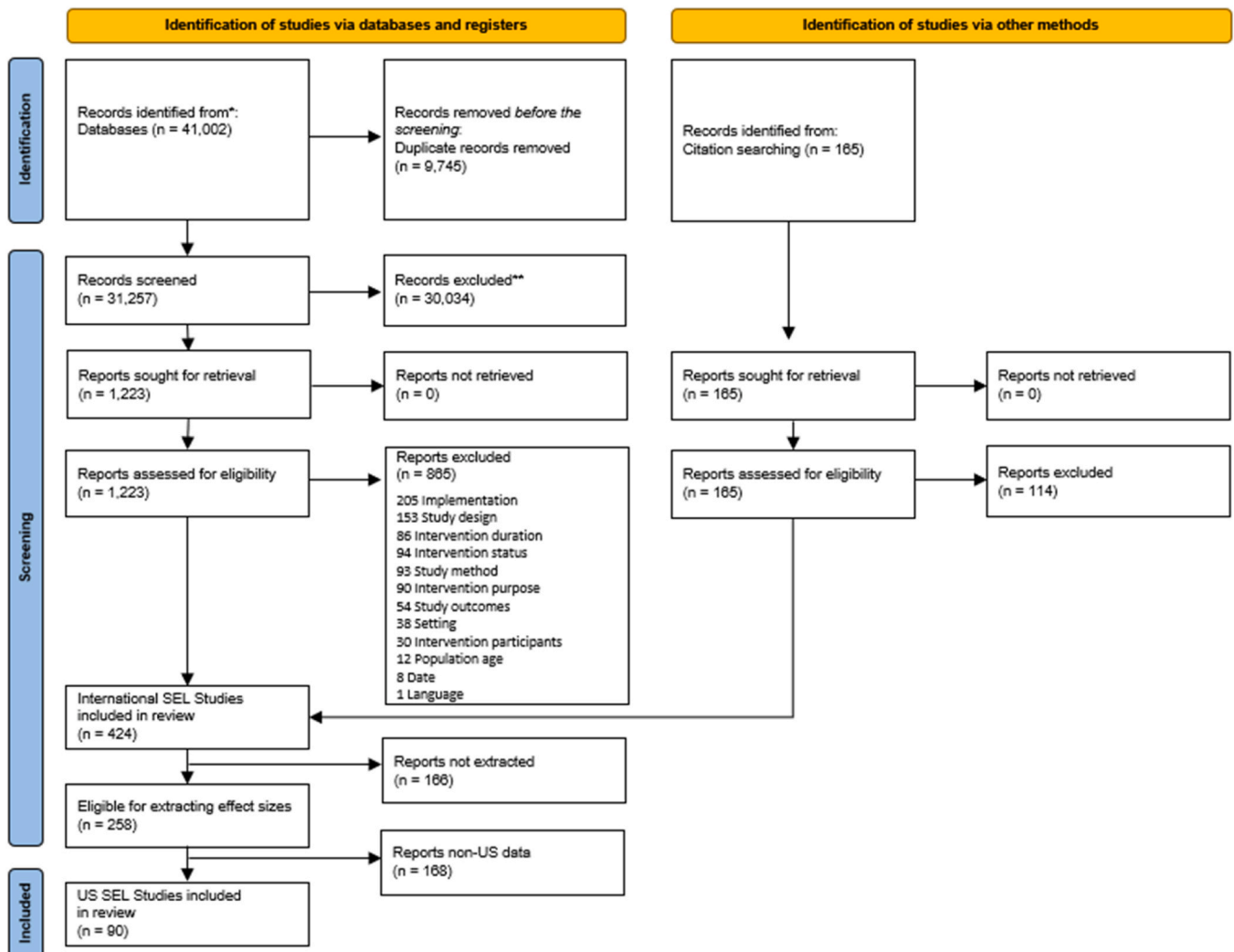


Fig. 1. PRISMA Flow Chart for Meta-analysis.

students or removed specific students from the classroom, were excluded. In the selection processes, studies ($N = 166$, 39.15%) were further excluded from the extant data set due to not reporting effect sizes at post-test for analysis. In the final screening for this study, we excluded studies conducted outside of the United States ($N = 168$, 65.12%). See PRISMA Fig. 1.

Coding procedure

Twelve coders (two undergraduates, four graduate students, and six authors coded articles using Qualtrics (see full codebook <https://osf.io/cszn8>). In order to effectively extract the large number of studies, and given the length of the codebook, three teams were established based on expertise to simultaneously extract study and student characteristics, intervention characteristics, and meta-analytic effects, respectively. All studies were double coded and given the heterogeneous nature of the body of literature on SEL, more than half the studies were triple coded to ensure reliability. To promote coder consistency, calibration exercises were conducted before the review and weekly throughout full-text coding, which lasted nine months. Discrepancies were resolved through discussion between coders, and a third coder arbitrated when needed. Kappa coefficients corrected for chance agreement were acceptable across all codes reported in this review (mean kappa was 0.89). Raters' agreements on continuous variables were strong, all above 0.90, and agreement on categorical variables ranged from .78–.92, respectively.

Student identity characteristics

To examine who is represented within the USB SEL empirical literature, extensive student identity characteristics were coded. Of note, codebook indicators had an 'other' open text option whenever possible, and coders manually cataloged the study description of an indicator for follow-up analysis. Student characteristics included (a) mean age, (b) age range, (c) developmental level (childhood, early adolescence, adolescence), (d) Socioeconomic Status (SES) reporting and (e) overall SES level, (f) percentage female, (g) reporting of LGBTQIA+ identification (i.e., sexuality-related minority status or related metric), (h) race/ethnicity, (i) language status, and (j) constructs (or any risk designation) related to disability. Furthermore, as an additional quality check, five of this study's authors re-reviewed the 90 United States studies that met our inclusion criteria for this analytic sample for additional student identifier information regarding student race, ethnicity, language, disability, gender, and sexual identity. No new data was extracted for analyses.

SEL outcomes

This meta-analysis included twelve dependent variable categories: (a) skills, (b) attitudes/beliefs, (c) prosocial behaviors, (d) externalizing behaviors, (e) civic attitudes and behaviors, (f) peer relationships, (g) emotional distress, (h) school functioning, (i) disciplinary outcomes, (j) school climate and safety, (k) family relationships, and (l) physical health.

In addition, in service of investigating the underpinnings of effective SEL programs, extensive intervention and study metrics were extracted. These indicators included (a) quality of study design, (b) intervention implementation fidelity, (c) intervention content (skills and competencies), specifically the number, combination and sequencing of said skills, (d) length of individual sessions and program, (e) delivery format (educators or non-educators), (f) theoretical underpinnings, (g) integration with academic instruction, and (h) SAFE consideration (sequenced, active, focused, explicit). Study quality, intervention content (skills and competencies) as well as number/combination and sequencing, and intervention characteristics (e.g., length, theoretical underpinnings, delivery format, SAFE, integration of the intervention into academic instruction, or intervention dosage and sequencing) were

also extracted and are reported in Cipriano, Strambler, et al. (2023).

Analytic strategy

Effect size calculations

We applied a precision-weighted mean effect size by Hedges's g formula (Hedges & Olkin, 2014; Lakens, 2013). As the weighted effect size of Cohen's d , the effect size from Hedges's g can provide more unbiased results, particularly when dealing with large samples (Cumming, 2012). For more accurate results, we assessed the impact of the USB SEL programs by comparing the intervention and control groups (e.g., using data results from Mean and SD, t -value, and F -value, etc.). Additionally, we calculated adjusted effect sizes, taking into account the baseline differences (Becker, 1988; Morris, 2008). The analytical methodology of this study aligns with the approach reviewed and conducted by Cipriano, Strambler, et al. (2023).

Analytic models

The analytic sample comprised 602 discrete SEL effect sizes derived from 90 distinct SEL studies, all conducted within the United States. In a priori power analysis, we referred to the parent SEL meta-analysis study result (Cipriano, Strambler, et al., 2023; overall $g = .194$) and evaluated the power of this meta-study using a calculator suggested by Quintana and Tiebel (2019). In the current study, the average number of participants in the effect size was greater than 600. For the conservative power test for this study, we set the expected effect size as .15, the average group size per study as 300, and the level of heterogeneity as large. The overall meta-analysis study power was close to .99 due to the large dataset ($k = 602$, $N = 90$). Even in the subdomain of peer relationship, which included the smallest number of cases ($k = 17$, $N = 10$), the calculated power level was .96.

Each SEL study included more than six mean effects in general. Therefore, in light of the clustered data, we opted for the final meta-analytic model that suitably handles the data structure by accounting for both correlation and hierarchical effects. To be more specific, we utilized the Correlated and Hierarchical Effects (CHE) model and embedded the Robust Variance Estimation (RVE) method as suggested by Pustejovsky and Tipton (2022). The results from the CHE and RVE methods consider the variance as an important component in synthesizing meta results within hierarchical data structures (Pustejovsky & Tipton, 2022). These analytical methods greatly enhance the accuracy of mean effect estimations in studies that report multiple effect sizes with varying outcomes. These analytic models were facilitated by the *clubSandwich* package in R (Pustejovsky & Tipton, 2022).

We executed the analysis models using the R software, with the packages of "metafor", "clubSandwich", and "robustmeta". To account for correlation both within and between studies, we conducted the pilot tests, and applied two-level clustering settings and set the ρ value at 0.60 for all the analyses (Cipriano, Strambler, et al., 2023). We assessed the mean effect, robust confidence intervals, p -values, and heterogeneity results, basing these evaluations on Hedges' g estimations. Furthermore, we tested the differences in SEL intervention effect by comparing program implementation features and gender subgroups. To answer this research question, we reselected the SEL intervention studies that reported effect sizes by different gender subgroups and reported the intervention effect using this sub-dataset. Lastly, we ran Egger's test to assess the potential publication bias. However, the 'metafor' package in R does not execute bias testing for a small number of studies (with minimum study count, $k_{min} = 10$). Therefore, the results we reported included cases that adhered to this minimum recommended observation count in each SEL domain and with reliable statistical power level as .80.

Publication bias

The analytic sample included 25 unpublished studies (e.g., dissertations, white papers, theses, technical reports, conference papers). Since published research often yields higher average effect sizes than unpublished work (Polanin et al., 2016), including both is crucial for a balanced view. To aid in identifying and reporting any potential publication bias, we followed the PRISMA framework’s methodological guidance. We examined publication bias across twelve SEL subdomain levels using Egger’s test (Egger et al., 1997; Sutton, 2009) which helped determine the statistical significance of any asymmetrical contributions among the studies we included (Pigott & Polanin, 2020). Furthermore, we used funnel plots, a graphical-based tool for publication bias, to identify any outliers or biased studies. These plots helped inform our decisions regarding methodological solutions to minimize bias (Ha, 2023) by identifying outliers that can negatively impact the results. As a result of this review, we adjusted two potential outlier effect sizes ($k = 2$) in our study to a maximum/minimum value at the $\pm 2 SD$ levels (Cipriano, Strambler, et al., 2023).

Results

Descriptive characteristics of reviewed studies

SEL programs

Characteristics of the 90 universal school-based SEL studies included in this review are summarized in Table 1. Our search for studies focused on the period between and inclusive to 2008 and 2020 with the range of 2008–2011 (22.2%), 2012–2015 (41.1%), 2016–2019 (27.8%), and 2020 (8.9%). Studies included 65 peer-reviewed published articles (72.2%), 21 unpublished theses (23.3%), and four technical reports/program evaluations (4.4%). Studies represent 55 dissemination outlets. Over half of the studies did not report a funding source (52.2%), and 37 studies were federally funded (41.1%), four reported private foundation information (4.4%) and only two studies reported being funded by state and through university grants, respectively. Studies consisted of 32 randomized control trials (35.6%) and 58 quasi-experimental designs (64.4%).

Studies reflect 59 unique USB SEL programs, 47 named and 12 unnamed. Thirty-nine of the SEL programs were represented once in our sample, while 13 SEL programs accounted for almost half of the total of studies (48.9%). These interventions, in order of greatest frequency, are *Strong Kids* ($N = 7$), *INSIGHTS* and *Second Step* ($N = 6$ each), *Positive Action* ($N = 5$), *Incredible Years* and *PATHS* ($N = 3$) *CW-FIT*, *Olweus*, *SEALS*, *SSIS-CIP*, *Strong Start*, *SSS*, and *Tools for Getting Along* ($N = 2$ each). Over half of the SEL programs ($N = 49$, 54.4%) in our review were based on an explicit theory, with the most frequently endorsed including social cognitive theory ($N = 10$), the theory of triadic influence ($N = 10$), temperament ($N = 8$), emotional intelligence ($N = 6$), and an ecological model ($N = 6$). The SEL skills that were most frequently endorsed include self-management ($N = 83$, 92.2%), cognitive regulation ($N = 79$, 87.8%), emotional processes ($N = 79$, 87.8%), social/interpersonal skills ($N = 76$, 84.4%), and responsible decision making ($N = 73$, 81.1%). See Table 2 for full distribution of SEL programs in this review.

Implementation features

Specific program characteristics and implementation features from the 59 USB SEL programs including the primary agent of change, effective program features, duration, and family and community engagement were extracted. Teachers and classroom aids primarily implemented programs ($N = 61$, 67.8%). Similarly, we found 59 (65.6%) studies had programs that endorsed SAFE (Sequenced, Active, Focused, Explicit) practices. Adult SEL development was explicitly a part of 16 studies (17.8%). Most programs spanned four months to one

Table 1
Descriptive Characteristics of the 90 Universal School-Based SEL Interventions.

General Study Features	N	%
Date of Report		
2008-2011	20	22.2
2012-2015	37	41.1
2016-2019	25	27.8
2020	8	8.9
Source of Report		
Published Article	65	72.2
Dissertation/Master's Thesis	21	23.3
Technical report, program evaluation	4	4.4
Dissemination Outlets	55	
Peer Reviewed		
Yes	65	72.2
No	25	27.8
Funding Source		
No funding indicated	47	52.2
Federal	37	41.1
Private foundation	4	4.4
State	1	1.1
University Grant	1	1.1
Study Design		
Randomized Control Trial	32	35.6
Quasi-Experimental Design	58	64.4
Reported Implementation Data		
Yes	52	57.8
Level of Implementation		
Satisfactory	3	3.3
Low	1	1.1
Moderate	6	6.7
High	28	31.1
Variable Across Sites	9	10.0
Sample Ecology	N	%
Age of students at Pre		
Elementary School (Kindergarten/ 5 yrs old – 5th grade/10 yrs old)	56	62.2
Middle School (6th grade / 11 yrs old-8th grade / 13 yrs old)	19	21.1
High School (9th grade / 14 yrs old 12th grade / 17 yrs old)	15	16.7
Types of Schools		
Unknown	45	50.0
Public	41	45.6
Public Charter School	2	2.2
Private: Religious Affiliation	2	2.2
School Setting		
Unknown	34	37.8
Urban, not otherwise clarified	19	21.1
Urban, inner-city, low-income area	10	11.1
Suburban	11	12.2
Rural	7	7.8
Combination of settings	9	10.0
Average School Size		
Unknown	63	70.0
Discrete SEL Programs		
Named	47	79.7
Unnamed	12	20.3
Programs reflected more than once in study	13	
Based on Theory	49	54.4
Primary Change Agent		
Teachers/aides	61	67.8
School staff other than teachers/aides (school mental health professionals)	7	8.0
Researcher and/or their staff	18	20.0
Community members or parents	3	3.3
Other/Unknown	1	1.1
Intervention Duration		
Up to half a school year (4 months)	43	47.8
Up to 1 whole school year	35	38.9
More than 1 whole school year	11	12.2
Unknown	1	1.1
Features of Effective Programs		
SAFE (Sequenced, Active, Focused, Explicit)	59	65.6
Free-Standing	76	84.4
Classroom-Based	88	97.8
Beyond the Lesson	48	53.3
Supports School Climate	57	63.3
Supports SEL Integration in Academics	27	30.0
Tiered Intervention	33	36.7

(continued on next page)

Table 1 (continued)

General Study Features	N	%
Multi-Phased Implementation	57	63.3
Multi-Component Intervention	40	44.4
Instructs Skills in an Order	78	86.7
Supports Adult SEL	16	17.8
Family Engagement		
No	41	45.6
Letters/emails sent home	38	42.2
HW	4	4.4
Explicit training+	7	7.8
Community Engagement		
No	78	86.7
Service-Learning projects	5	5.6
Community volunteers+	7	7.8
Steps to increase program implementation		
Pre-program training was provided	5	5.7
Ongoing, support or training was provided once the program began	1	1.1
Detailed program manuals or protocols provided to guide providers	2	2.3
Combination of above (list)	68	77.3
Other	1	1.1
None of these steps or strategies	9	10.2
Unknown	2	2.3
SEL Skills and Strategies Reflected in Interventions		
Two Domain Framework	90	100.00
Interpersonal Skills	76	84.4
Intrapersonal Skills	74	82.2
CASEL 5 Competency Framework	89	98.9
Self-Awareness	71	79.6
Self-Management	83	92.2
Social Awareness	56	62.2
Relationship Skills	67	74.4
Responsible Decision Making	73	81.1
Framework for Social and Emotional Learning	86	97.7
EASEL Taxonomy	89	98.9
Cognitive Regulation	79	87.8
Emotional Processes	79	87.8
Social/Interpersonal Skills	76	84.4
Values	45	50.0
Perspectives	29	32.2
Identity	30	33.3

school year ($N = 78, 86.7\%$). Regarding family engagement within the programs, four studies involved families through homework (4.4%) and seven involved explicit familial training (7.8%). Furthermore, most studies did not have an explicit community engagement component ($N = 78, 86.7\%$). Just over half of studies ($57.8\%, N = 52$) reported any implementation data for analysis, with less than a third ($31.1\%, N = 28$) reporting the implementation level as high. Aspects of implementation that were assessed include fidelity ($61.1\%, N = 55$), dosage ($43.3\%, N = 39$), quality ($35.6\%, N = 32$), and student engagement or responsiveness ($32.2\%, N = 29$). Program implementation was most frequently assessed using a combination of self-reports, behavioral observations, and recorded sessions. Studies endorsed implementing various steps to increase program implementation, including providing pre-program training, providing ongoing support once the program began, and providing detailed program manuals or protocols. See Table 1 for full frequencies of implementation features.

Additionally, regarding characteristics of school settings and types in the reviewed sample, only half the sample reported where they were conducted, with most being conducted in public schools ($N = 41, 45.6\%$). Similarly, 34 (37.8%) studies did not report if they were conducted in an urban, suburban, or rural setting; of those that did, 29 were urban (32.2%), 11 were suburban (12.2%), and 7 were rural (7.8%) (See Table 1).

Student demographics

Detailed student demographic characteristics were extracted including also the presence, absence, and use of sample level data.

Table 2

Frequencies of Named SEL Interventions Represented in Study.

SEL Intervention Name	f	Citation
Anti-Bullying Initiative	1	Beightol, 2012
Behavioral, Academic, and Social Engagement (BASE)	1	Dawes, 2020
Building Assets, Reducing Risks (BARR)	1	Bos et al., 2019
Capturing Kids' Hearts	1	Holtzapple et al., 2011
Class-Wide Function-Related Intervention Teams (CW-FIT)	2	Caldarella et al., 2015;Wills et al., 2016
Connect with Kids	1	Mogro-Wilson & Tredinnick, 2020
COPE Healthy Lifestyles TEEN	1	Melnik et al., 2009
Critical Civic Inquiry	1	Hipolito-Delgado & Zion, 2017
Dream It! A Playbook to Spark Your Awesomeness	1	Jacquez et al., 2020
Emotional ABCs Program	1	Ehret, 2020
Growing Early Mindsets	1	Coates, 2016
Growing Minds	1	Willenbrink, 2018
I Can Problem Solve (ICPS)	1	Boyle & Hassett-Walker, 2008
Incredible Years	3	Webster-Stratton et al., 2008;Murray et al., 2018;Chuang et al., 2020
INSIGHTS	6	McClowry et al., 2010;O'Connor et al., 2014;McCormick et al., 2015c; McCormick et al., 2015c;McCormick et al., 2016;McCormick et al., 2019
It's Your Game. Keep It Real (IYG)	1	Peskin, 2014
Journey of Hope	1	Powell, Bui, (016
Learning to BREATHE	1	Felver et al., 2019
Lessons in Character	1	Hanson et al., 2012
Master Mind	1	Parker et al., 2014
MBSEL	1	Bungum, 2014
Olweus Bullying Prevention Program	2	Stone, 2009;Losey, 2009
PATHS	3	Crean & Johnson, 2013;Fishbein et al., 2016;Calhoun et al., 2020
PAX Good Behavior Game	1	Weis 2015
Positive Action	5	Beets et al., 2009; Washburn 2010; Snyder et al., 2012;Lewis et al., 2013; Silverthorn et al., 2017
Project Team	1	Street, 2013
RULER	1	Brackett et al., 2012
SEALS	2	Farmer et al., 2010;Hamm et al., 2014
Second Step	6	Jack et al., 2009; Low et al., 2015; Espelage 2015; Espelage 2015;Espelage et al., 2016;Benson, 2017
Social Skills Improvement System Classwide Intervention Program (SSIS-CIP)	2	Diperna 2016; DiPerna 2018
Steps to Respect (STR)	1	Low et al., 2010
Strong Kids	7	Guedner, 2008;Guedner & Merrell, 2011;Kramer, 2013;Kramer et al., 2014; Cook et al., 2015;Skiba, 2017;Hetrick (2018)
Strong Start	2	Caldarella et al., 2009;Fewkes, 2017
Strong Teens	1	Castro-Olivo, 2014
Student Success Skills (SSS)	2	Lemberger-Truelove et al., 2015;Kane, 2015
Take the Lead (TTL)	1	Domino, 2013
TestEdge	1	Donato, 2009
The 4Rs Program	1	Jones et al., 2010
The Tai Chi Tiger Program	1	Wright et al., 2010
Tools for Getting Along	2	Daunic et al., 2012;Daunic et al., 2019
Tools for Life	1	Gonzales et al., 2020
Tools of the Mind	1	Blair et al., 2018
Transformative Life Skills (TLS)	1	Frank et al., 2017
Violence Prevention project	1	Thompkins et al., 2014
Williams LifeSkills® (WLS)	1	Barnes et al., 2012
Yoga Ed	1	Khalsa et al., 2012
Zippy's Friends	1	Rodker, 2013
Unnamed	12	Ho et al., 2011;Kokinakis, 2011; Coleman, 2012;Hagins et al., 2013; Kanagy-Borofka, 2013;Britton et al., 2014;Daly et al., 2015; Viafora et al., 2014;Butzer et al., 2017;Meadows, 2018; Meyer, 2018;Johnstone et al., 2020

Age

All 90 studies in the sample reported student age. More than half of the studies included students in elementary school (K-5th grade; $N = 56$, 62.2%) while 19 studies included middle school students (6th –8th grade) and 15 included students from high school (9th –12th grade).

Gender and sexuality

Eighty studies reported student gender (88.9%; see Table 1), and all represented gender using a binary of Female/Male. Sexual orientation was not reported in any of the 90 U.S. studies. Seventy nine studies (84.4%) included the proportion of each gender subgroup in the sample.

Race and ethnicity

Sample level student race/ethnicity was not reported for 23 studies (25.6%). Within the remaining 67 studies, race/ethnicity was reported as American Indian or Alaskan Native ($N = 13$), Asian ($N = 27$), Black or African American ($N = 58$), Hispanic/Latino(a) ($N = 50$), Native Hawaiian or Asian/Pacific Islander ($N = 8$), more than one race ($N = 9$), and white ($N = 50$). Additional specific racial and ethnic identifiers included: Cuban, East Indian, Filipino, Multi-Hispanic, and Puerto Rican. Examples of generalizations as student identifiers included: “Minority students,” “other Hispanic,” “other,” and “multirace” (see Table 3). Race or ethnicity was explored as a covariate in 67 studies (74.4%).

Table 3
Student Identity Representation.

Gender Representation	<i>N</i>	%
Mentioned Student Gender	80	89.9
Included Proportion of Each Gender Subgroup in Sample	76	84.4
Disability Representation	<i>N</i>	%
Any Disability Status	30	33.3
At Risk for Emotional or Behavioral Disorder (EBD)	5	1.1
Attention Deficit Hyperactivity Disorder (ADHD)	2	2.2
Autism Spectrum Disorder and ADHD	1	1.1
Behavioral Problems	1	5.6
Cross-Categorical Special Education	1	1.1
Emotional Disturbances	1	1.1
Exceptional Student Educational Services	1	1.1
Individual Education Plan (IEP)	6	4.5
Multiple Disability	1	1.1
Special Education Services/SPED-Eligible/SEND	15	16.7
Speech Impairment	3	3.3
Students with Disabilities (SWD)	1	1.1
<i>Additional Disability representations captured in the review</i> Emotional/Behavioral/Learning Disorder, Special needs		
Explored Disability as a Covariate	3	3.3
Race and Ethnicity Representation	<i>N</i>	%
Any Race/Ethnic Identifier	67	74.7
American Indian or Alaskan Native	13	14.4
Asian	27	30.0
Black or African American	58	64.4
Hispanic/Latino(a)	50	55.6
Native Hawaiian or Asian/Pacific Islander	8	8.9
More than one race	9	10.0
white	50	55.6
Unspecified	14	15.6
<i>Additional Ethnic and Racial Identifiers captured in review</i>		
Specific: Cuban, East Indian, Filipino, Multi-Hispanic, Puerto Rican		
Generalizations: Minority students, other Hispanic, other, multirace		
Explored Race or Ethnicity as a Covariate	67	74.4
Language Status Representation	<i>N</i>	%
English Language Learner Services	13	14.4
Immigration Status	2	2.2
Parent Language Status	1	1.1
<i>Additional Linguistic Identifiers captured in review</i>		
Number of languages spoken, Parent education level, Non-dominant language speakers, Primary English speakers, English fluency		
Explored Language Status as a Covariate	0	0

Notes. In discussing the intervention results for the three demographic groups, studies used each group as a control variable (i.e., covariate) and did not provide detailed descriptive results for each demographic subgroup.

Language status

Student language status was represented in 16 studies and was indicated primarily by *English Language Learner Services* ($N = 13$), as well as *Immigration Status* ($N = 2$), and *Parent Language Status* ($N = 1$). Additional miscellaneous language status identifiers coded included: “Number of languages spoken,” “Parent education level,” “non-dominant language speakers,” “Primary English speakers,” and “English fluency.” No studies looked at the differential effects of SEL programs by student language status.

Students with disabilities

Among all included studies of SEL conducted in the U.S. from which we could calculate effect sizes ($k = 90$), 30 studies (33.3%) reported on participants’ demographic information related to disability and within these studies, half of them primarily represented students with disabilities by indicating students received *Special Education Services/SPED-Eligible/SEND* ($N = 15$, 50%). Three studies (3.3%) explored disability as a covariate.

Do SEL programs significantly improve students’ skills, attitudes, and behaviors in the United States?

We assessed the mean effects of the SEL programs in the U.S. to determine whether the SEL programs significantly improve K-12 students’ outcomes across twelve SEL domains. Table 4 shows the mean effect size between the pre- and post-test of the SEL programs in the United States educational contexts. We followed PRISMA guidelines and our selection criteria to examine 90 empirical studies, which covered 602 cases across all SEL outcome domains. The Q-value of the overall SEL outcomes was significant, indicating heterogeneity among the included SEL intervention effects ($Q [601] = 8194.530, p < .001$). To account for heterogeneity, we used the *Random Effects* (RE) model to analyze the data considering the clustered structure of SEL empirical studies. Table 4 shows the results of the mean effect of SEL program across the different outcome domains. The overall mean effect of the SEL program was .193 and statistically significant between the intervention and control groups ($k = 602, g = .193, 95CI = [.141;.244], p < .001$). The meta-analysis results support that SEL programs in the U.S. contributed to improving K-12 students’ SEL outcomes in school contexts.

Furthermore, we estimated the mean effects of the SEL programs implemented in the U.S. by twelve SEL outcome domains. Nine domains (i.e., *SEL skills, attitudes/beliefs, prosocial behaviors, externalizing behaviors, peer relationship, emotional distress, school functioning, school climate/safety, and physical health*) that met the recommendable number of samples (i.e., $k \text{ min.} = 10$) for reporting reliable results with the publication bias test (Valentine et al., 2010). Effect sizes of SEL programs in seven among the nine domains were statistically significant, presented here at the domain level in order of greatest magnitude. Students who participated in the SEL programs in the U.S. experienced significant positive changes in *school climate/safety* ($k = 47, g = .319, 95CI = [.087, .552], p = .012$), *attitudes/beliefs* ($k = 49, g = .183, 95CI = [.092;.274], p = .001$), *prosocial behaviors* ($k = 64, g = .178, 95CI = [.053;.303], p = .007$), *SEL skills* ($k = 102, g = .171, 95CI = [.098;.245], p < .001$), and *school functioning* ($k = 90, g = .096, 95CI = [.032;.161], p = .005$), and significant reductions in *externalizing behaviors* ($k = 120, g = -.152, 95CI = [.080;.224], p < .001$), and *emotional distress* ($k = 77, g = -.120, 95CI = [.056;.184], p = .001$). The mean effects of the other two domains, *peer relationship* ($k = 17, p = .010$) and *physical health* ($k = 24, p = .058$), were not statistically significant but were in a marginalized range of p -value under .10 (see Table 4).

Are the effects of USB SEL programs maintained after the program ends?

In this study, the selected SEL empirical studies included at least 16 weeks (i.e., one semester) or a minimum of six lessons. Following these

Table 4Meta-Analytic Results of Mean Effect Sizes (Hedges' g) and 95% Confidence Intervals at Post-Intervention between Intervention and Control Conditions (Robust Random-effect Model by Study).

Domain	Overall	School Climate/ Safety	Attitudes/ Beliefs	Prosocial Behaviors	SEL Skills	Externalizing Behaviors	Emotional Distress	School Functioning	Physical Health	Peer Relationship
Total Sample	0.193	0.319	0.183	0.178	0.171	0.152	0.120	0.096	0.307	0.159
with	[.141,.244]	[.087,.552]	[.092,.274]	[.053,.303]	[.098,.245]	[.080,.224]	[.056,.184]	[.032,.161]	[-.018,.631]	[-.038,.355]
paper	< .001***	.012*	.001**	.007**	< .001***	< .001***	.001**	.005**	.058*	.100*
k	602	47	49	64	102	120	77	90	24	17
N	90	11	20	25	39	40	33	37	6	10
Q	Q (601)	Q (46)	Q (48)	Q (63)	Q (101)	Q (119)	Q (76)	Q (89)	Q (3)	Q (16)
(df)	= 8194.530***	= 792.934***	= 324.954***	= 467.859***	= 1192.845***	= 989.046***	= 145.013***	= 405.835***	= 1447.815***	= 93.196***
Eggers' test	t (600) =	t (45) =	t (47) =	t (62) =	t (100) =	t (118) =	t (75) =	t (88) =	t (22) =	t (15) =
(df)	.55,	7.82,	-1.10,	1.47,	.62,	1.36,	1.15,	1.39,	-.30,	1.32,
p	.585	< .001***	.276	.148	.538	.177	.255	.167	.766	.208

Notes. To estimate the overall effect size across all outcome domains, this table reported results in two domains (Externalizing behaviors and Emotional distress) with converted positive scores of ES (Hedges' g). The outcome 'Academic Achievement' was also evaluated as a subset of the domain (School Functioning) $g = .082$; 95% CI = [.009,.155]; $p = .029$ *)

* $p < .05$, ** $p < .01$, *** $p < .001$, and marginal $p < .10$.

inclusion criteria, to precisely analyze the follow-up effect sizes of these programs, we examined whether the differences in SEL results between the two groups persisted after a lag period of one semester (six months), equivalent or longer to the length of the intervention. We identified 38 follow-up effects from four SEL programs ($k = 38$, $N = 4$; *INSIGHTS*, *Tools for Getting Along*, *Tools of the Mind*, *Yoga Intervention*) and assessed the effects between the initial and follow-up measurements. In this study, the overall follow-up effect was not statistically significant ($k = 38$, $g = .041$, 95% CI = [-.058,.140], $p > .05$). At the subdomain level, our analytic samples did not reach a reliable statistical power or a minimum effect sample for testing sample bias (i.e., $k \min. = 10$).

Test of publication bias

Fig. 2 shows the publication bias results of this study with funnel plots. In the overall sample of the SEL effects ($k = 602$, $N = 90$) in twelve SEL domains in the United States, Egger's test result was not significant (t [600] = 0.550, $p = 0.585$), indicating no concern of publication bias. In this meta-analysis study, we identified nine of twelve SEL outcome domains applicable (i.e., $k \min. = 10$) for testing the publication bias (e.g., *SEL skills*, *attitudes/beliefs*, *prosocial behaviors*, *externalizing behaviors*, *peer relationship*, *emotional distress*, *school functioning*, *school climate/safety*, and *physical health*). Table 2 details there was no serious concern of publication bias in six domains between the pre- and post-intervention, and that there was minor concern of caution for *school climate/safety* due to the distribution of studies showing no normal curve (t [45] = 7.82, $p < .001$), (see Table 2). Additionally, we ran Egger's test for the follow-up effects of overall SEL outcomes. The statistical result was estimated from a small sample ($k = 38$, $N = 4$) and not significant (t [36] = -0.550, $p = .584$).

Do the effects of SEL programs in the U.S. significantly differ when they are implemented by the classroom teacher or at different levels of school?

We additionally evaluated the differences of SEL program effect by

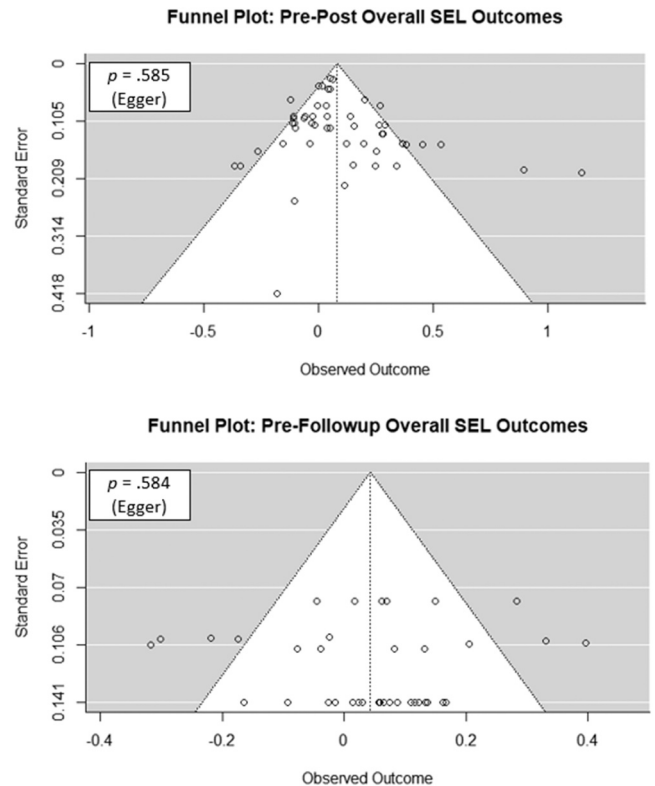


Fig. 2. Funnel Plots of Overall SEL Outcomes (Pre-Post versus. Pre-Followup).

different implementation settings, including who implemented the program and the age group the program was serving. We compared the intervention effect by if a program was delivered by the teacher as compared to those programs delivered by someone other than the teacher (e.g., school mental health staff, researcher, or program provider) and also compared the effects of SEL programs on student outcomes by level of schooling where the program was implemented (Elementary v. Secondary), respectively.

First, we identified 451 effects derived from 64 empirical studies (reflecting 40 SEL programs) delivered by teachers and 187 effects derived from 29 empirical studies (reflecting 23 SEL programs) delivered by someone other than the teacher. The moderation effect of teacher SEL instruction was statistically significant after the hierarchical clustering setting in the CHE model ($b = .107$, $se = .047$, $95CI = [.012, .201]$, $p = .028$); students who participated in SEL programs facilitated by their teacher had more significant outcomes compared to students who participated in SEL programs delivered by someone else other than their teacher. Next, we compared the overall effects of the 394 effects in elementary education settings with the 208 effects in secondary education settings. The moderation test results were not significant by the different school levels ($b = -.042$, $se = .055$, $95CI = [-.153, .069]$, $p = .450$), indicating there were no significant differences between elementary and secondary students and that the impact of SEL programs on student outcomes was equally effective for students in both elementary and secondary school programs levels.

Do effects of SEL programs in the U.S. significantly differ as a function of student identity?

We present the meta-analysis results of the SEL intervention effect categorized by gender subgroups. For subgroup analyses of race, ethnicity, and disability variables, the current sample was not sufficient for results with a statistical power level of .80; therefore, we reported descriptive results based on the systematic review.

Gender

We identified four studies ($N = 4$; $k = 28$; Beightol et al., 2012, Benson, 2017; Parker, Kupersmidt, Mathis, Scull, Sims, 2014, & Weis et al., 2015) from the dataset that reported results on the effects of SEL implementations by gender subgroups between the pre and post-test. All four empirical studies reported the gender information of students as binary (male/female). The 28 extracted effect size row data ($k = 28$) included the same constructs from females ($k = 14$) and males ($k = 14$). First, the random effect model analysis of this sub-dataset showed that the overall SEL intervention effect was not statistically significant ($k = 24$, $g = 0.135$, $95CI = [-.216, .486]$, $p = .262$). Based on the sub-sample results, we additionally ran a meta-analysis in each gender subgroup. In both gender subgroups, the effect of the SEL intervention did not significantly differ between the treatment and control groups in both the male group ($k = 14$, $g = 0.092$, $95CI = [-.170, .353]$, $p = .559$) and female group ($k = 14$, $g = 0.128$, $95CI = [-.041, .327]$, $p = .062$). The moderation results of the meta-analysis with the sub-sample (e.g., including gender subgroup data) were not significant for the different gender groups ($k = 28$, $g = 0.014$, $95CI = [-.455, .483]$, $p = .877$).

Race and ethnicity

We identified 67 studies that reported the race and/or ethnicity of student participants as a covariate in their analyses. For examples of the range of ways race and ethnicity were applied as covariates across analyses of SEL program effects (see Jones et al.'s, 2010 evaluation of 4Rs, Frank et al.'s, 2016 evaluation of *Transformative Life Skills*, DiPerna et al., 2016 evaluation of *Social Skills Improvement System Classwide Intervention Program* (SSIS-CIP), and McCormick et al., 2015b evaluation of *INSIGHTS*).

Descriptive analyses provide insight into how race and ethnicity may impact and/or moderate intervention effects. For example, we identified one SEL program, *Second Step*, with two studies that controlled for race but examined overall differences in outcomes between non-African American students and African American students (Espelage, Low, et al., 2015; Espelage, Rose, et al., 2015). In examining bully perpetration, victimization, and physical aggression pertaining to students with disabilities in schools with and without *Second Step*, Espelage, Rose, et al. (2015) reported a significant decrease in perpetration in intervention schools. When examining differences based on race and ethnicity status, the authors found that bullying perpetration was more likely to be endorsed by white students, bullying victimization was endorsed less often among African American students compared to Hispanic students, and physical aggression was endorsed at higher rates among white students. Further, Espelage, Low, et al. (2015) examined intervention effects across seven outcome variables among schools across two states, Illinois and Kansas, and found that reports of sexual harassment and violence perpetration and being the target of homophobic name calling were significantly less likely in Illinois *Second Step* schools. Again, the authors controlled for race and explored the overall differences between African-American students and students of other racial and ethnic backgrounds. Espelage, Low, et al. (2015) found differences in endorsement levels for physical aggression between African American students and Asian, Biracial, Hispanic, and white students while differences were found between Hispanic students and African American students regarding endorsements of peer victimization. The authors also reported significant differences in levels of endorsements between African American students and Asian, white, and Hispanic students regarding physical aggression, homophobic name calling and victimization, and sexual violence and victimization.

Furthermore, Pekin et al., (2014) explored effects for middle school students who participated in *It's Your Game...Keep it Real*, an intervention targeting dating violence. Their analyses found that students participating in the intervention were less likely to report physical and emotional dating violence victimization and emotional dating violence perpetration compared to students in the control group. The authors further stratified the intervention effects by race and ethnicity status, discovering that Hispanic students in the treatment group had lower levels of emotional dating violence victimization and perpetration after the intervention, while African American students saw a decrease of physical dating violence victimization.

In addition, studies specifically examining moderation effects of an intervention based on race and ethnicity status also reveal differences across student identities. Dawes et al. (2020) explored moderation effects of race and ethnicity status among schools with and without the *Behavioral, Academic, and Social Engagement* (BASE) classroom management model across nine outcomes. The authors found no program effects across all outcomes but did discover moderation effects based on race and ethnicity status among two outcomes. More specifically, racially marginalized youth were more protective of bullied students and showed a reduction in defiant behavior. Likewise, Wike (2011) examined the impact *Making Choices*, an intervention focusing on social-emotional skills, had on fourth and fifth grade students' social competence, emotion regulation and prosocial and aggressive behavior, and found moderating effects of race and ethnicity status. Students in fifth grade had lower levels of relationship aggression and improved emotion regulation, students in fourth grade reported lower levels of overt aggression, and students across both grades had improved prosocial behavior and social competence. Wike (2011) also discovered that fourth grade African American students saw significantly more improvements on teacher-reported overt and relational aggression.

Given the lack of consistency in how race and ethnicity status are explored and reported on in SEL intervention research, it is not possible to meta-analytically understand how student racial and ethnic identity may impact intervention effects. However, these descriptive findings across these studies suggest that reporting general results that explore

only differences between treatment and control groups may mask differences that exist among individual outcomes based on race and ethnicity status, and suggest over and above benefits for students of color from SEL programs.

Students with disabilities

Three studies reported the effects of SEL programs on students with disabilities. DiPerna et al. (2018) found no significant differences in the impact of *Social Skills Improvement System Classwide Intervention Program (SSIS-CIP)* between students with and without disabilities and Hetrick (2018) reported no differences in outcomes between students with and without special education support who participated in *Strong Kids*. Unlike DiPerna et al. (2018) and Hetrick (2018), Kramer et al. (2014) examined the effectiveness of SEL on a specific group of students with disabilities, students with emotional and behavioral disorders (EBD), and found that after students participated in *Strong Kids*, internalizing symptoms were significantly lower in students with EBD than in those without EBD.

Discussion

The goal of this study was to review the evidence for universal SEL programming within the United States to understand nation-specific trends in SEL programming and provide specific recommendations in consideration of marginalized students in the U.S. To accomplish this goal, this paper systematically analyzed the evidence of effectiveness of universal SEL programs from 2008 through 2020. Upon completion of analyses, we report robust evidence for the effectiveness of SEL programs on student academic, social, and emotional as an aggregate, nation-specific trends in SEL programming comparative to the previously analyzed international sample, and opportunities for specific and differential effectiveness of studies for students with marginalized identities within the United States literature. We attend to these findings in turn here and conclude with promising future directions to expand how we understand the impact of SEL programs in the U.S.

SEL programs in the United States promote positive academic, social, and emotional outcomes for students across K-12

The present study documents the significant and positive effects of SEL programs in the U.S. on a range of student outcomes, including increased student perceptions of school climate and safety, SEL skills, attitudes and beliefs, prosocial behaviors, and school functioning, and reductions in students externalizing behaviors and experiences of emotional distress. Findings extend those originally reported by Durlak et al. (2011) and unsurprisingly given where these data were drawn, mirror nearly all those reported by Cipriano, Strambler, et al. (2023) in the international data set used for these analyses. Notably, the U.S. SEL studies in this review showed a less biased normal distribution, including unpublished works comprising 28% of the data. Considering the nature of the present data structure is more homogeneous than our prior review, the derived results from the U.S. SEL programs provide more precise results from an analytic perspective on the effects of SEL programs exclusively in the U.S. context.

Teachers are the strongest implementers of SEL programs in the United States

Student outcomes are strongest when the SEL program is delivered by their teacher. Findings point to the importance of training pre- and in-service educators to optimize SEL implementation for the benefit of all students in the United States. Teachers' social and emotional competencies are vital for SEL implementation and influence the efficacy of SEL programs (Schonert-Reichl, 2017). This area of need is particularly critical in the United States. We note that in our review more than 80%

of studies did not incorporate an adult SEL component during the implementation of their respective SEL programs and currently there is no standardized certification requirements in SEL for teachers K-12 (Hon et al., 2024). There is much room for development in how teachers are supported to optimize SEL implementation for students. Furthermore, burgeoning research in the United States reporting the significant role of racial and ethnic congruences between teacher and student in promoting effective student outcomes warrants additional inquiry (Joshi et al., 2018; Oates, 2003; Wright, 2015). In 2020–2021, 80% of public school teachers in the United States were white compared to 46% of public school students (National Center for Education Statistics, 2023). As we increase the reporting of evidence for SEL, it would be helpful to concurrently consider how characteristics of the implementer (e.g., teacher's identity) intersect with those of their students to examine how to optimize the effects for students from SEL. Further investigation of the application of best practice in SEL implementation practices is warranted.

SEL programs in the United States benefit students in elementary and secondary school

Analyses revealed that SEL programs in the U.S. are equally efficacious for students in elementary and secondary school, respectively. This finding comes at a critical time in U.S. education policy as SEL is increasingly incorporated into state standards nationwide. While all 50 states have public education SEL standards for pre-K students, only 27 states have SEL standards for all students across K-12 (Stanford & Meisner, 2023). Our analyses indicate the efficacy of SEL programming across K-12, supporting the explicit inclusion of SEL programming and the adoption of SEL standards for all grade levels in the U.S.

SEL programs in the United States are varied

This systematic review of the effectiveness of 59 discrete SEL programs (47 named, and 12 unnamed) from a corpus of 90 studies illustrate the significant variability in the components and implementation of SEL programs. Although all SEL programs endorsed at least two of the CASEL competencies in the five-factor framework, programs did so in very different ways; SEL programs included in the review were based on 55 different theories and focused on different components of SEL competencies, skills, and strategies as a result. Furthermore, aspects of implementation varied as well; session duration ranged from 10 min to 120 min per session, and overall program duration ranged from 4 months to more than one school year. Although this heterogeneity was found to be beneficial as an aggregate in support of student academic, social, and emotional outcomes, we encourage SEL programs in the United States to increase the specificity in how they are described to support informed decision making by schools, districts, and states regarding which SEL program to implement (Cipriano, Strambler, et al., 2023; Cipriano & Strambler, 2024; Jones et al., 2022).

SEL programs in the United States have limited engagement with families and communities

Our review further found limited engagement between SEL programs, families, and communities. Most studies did not involve the community at all ($N = 78, 86.7\%$), and SEL programs demonstrated very limited engagement with families through homework (4.4%), nor training (7.8%). Although these results mirrored those reported in the international meta-analysis (Cipriano, Strambler, et al., 2023), these findings have particularly salient ramifications for the U.S. Once not involved in educational political discourse and policy, families, and specifically parents, are increasingly playing an active role in shaping school climate and functioning as well as what their children learn and how educators can teach within the classroom. Over the last two years, parents and educators have held favorable views of SEL in public

education (CFC, 2022; McGraw Hill, 2021; PTA, 2022). Yet, currently organized parents' rights political movements are posing a threat to SEL (*Moms for Liberty*, 2022). Organized movements like this one are having rippling effects nationwide. Teachers have already noted that current restrictions on how they can teach, particularly around topics of race and gender, are a result of emerging parent involvement in and criticisms of teaching materials (Woo et al., 2023).

Perhaps the absence of families and communities within current SEL programming and practice in the evolving context in the U.S. is putting SEL at risk of becoming disconnected from the very stakeholders it needs to continue to grow and thrive. Although the politicization of SEL is largely a new debate among U.S. politics, it is possible that increased engagement with families and communities could help mitigate inaccurate interpretation of SEL findings and combat the false narrative that SEL is harmful. In addition, SEL programming that is driven by strong parent, family, and community partnerships may address and resolve any concerns that parents may have pertaining to SEL. Families, parents, and communities can be assets and a resource, especially as marginalized identities are incorporated into the design, implementation, and research of SEL programming and practice (See Cipriano & McCarthy, 2023 for further recommendations pertaining to fostering the family and community partnership when designing and implementing inclusive SEL programming).

SEL programs in the United States did not show evidence of an overall follow-up effect

In the follow-up effect analysis, we found notable differences between studies from the U.S. and those conducted internationally. We examined the follow-up effects from programs from at least 6 months after completing the SEL program. In the international sample, there were 43 studies that met this criteria for analysis. According to the results, the international studies showed significant effect after 6 months, which was not observed in the present analysis of exclusively U.S. studies. This result highlights the importance of evaluating SEL programs in the U.S. after the program ends. With only 4% of studies providing this data for analysis, it still remains largely unclear how students in the U.S. exclusively are benefiting from SEL programs in the long term. Given the precedence set by Durlak and colleagues in their 2011 paper, and the evidence of significance we found in our international analyses most recently, we suggest interpretation of these results with caution and encourage researchers to intentionally evaluate and report on the follow-up effects of programs to support the future of SEL science and practice to draw from these results analytically with a more robust and representative sample. The ability to collect and evaluate experimental design follow-up effects in the United States is riddled with barriers to execution, and will require explicit support from funding agencies to ensure resources necessary to support long-term evaluations in our nation's schools.

Some marginalized students benefit from SEL in the U.S.- and – we can do better

Our findings suggest evidence of benefits for some marginalized students in the U.S. and the possibility of recalibrating the way we do research to analytically support more inclusion of marginalized students in experimental research designs in the United States. The 90 studies systematically analyzed in this paper provided *some* representation of student identities, with nearly all presenting on student gender, nearly three-fourths mentioned student race, a third mentioning students with disabilities, and less than one in five mentioning student language status. Unfortunately, these data are presented in ways that limit the execution of meta-analytic analyses centering marginalized student experiences. On the surface, this finding is unsurprising; our prior work has documented both the levels of missingness (Cipriano, Naples, et al., 2024) and vast heterogeneity in student identity representation in

experimental studies of SEL (Cipriano & McCarthy, 2023). However, if all students in all samples are accounted for, we can move towards more consistent and specific representation across studies that will support opportunities to meaningfully aggregate subgroups to compare and understand if there are differential effects of SEL programs for students with marginalized identities. A primary recommendation stemming from this review is to increase open science practices including transparent and shareable data across SEL evaluations to provide further research opportunities for disentangling the heterogeneity of SEL program effects for who under what conditions.

Why do the results point to open science? Importantly, assumptions of adequate sample size to have the power to detect statistically significant effects will likely not be met exclusively by shifts in sample representation. Although increasing within-school/study sample representation will help, it is likely that the heterogeneity of student identities represented will not simply be adequately sized to detect subgroup effects. This is especially where data sharing and open science practices can benefit the SEL field. Students who are a numerical minority in their classroom and school, even with representation, may still be represented at a level that is too small to detect a subgroup effect. Consider how a third of studies mentioned students with disabilities, with half of them representing disability as a global indicator (*Special Education Services/SPED-Eligible/SEND*). Similarly, nearly all (84%) of the 18% of studies that mentioned language status used the framing of English learner, and no studies examined the differential effects of SEL for students with non-English language identities. Even if we continue and increase the use of the most generalized categories for student representation, we will only get so far in our efforts to understand if there are differential effects for students. Improving representation requires both differentiation and meaningful representation. The implications of these simplified standard categories can be egregious, as they can undermine both the distinct differences within categorical identity and promote the aggregation of identities across people in ways that diminish unique variations in their experiences of school (Cipriano, Naples et al., 2023; Cipriano & McCarthy, 2023).

As an illustration, let's consider the aggregation of African American and Black in SEL research in the United States. The identifier *African American* refers to an ethnic group of people that share a common ancestry rooted in the Atlantic Slave Trade (McGhee, 2022) whereas the identifier *Black* is a racial category that was created to suppress the human rights of those black and brown skin in the United States (Crenshaw, 1991; Leonardo, 2004; McGhee, 2022). Despite this distinction, African American has largely been considered the politically correct terminology for a population of people who are non-white presenting, potentially obscuring meaningful in-group distinctions and legacies of racism that overlook the rich diversity of ethnic representation in the U.S. (Cipriano & Naples, Eveleigh, et al., 2023; Leonardo, 2004; McGhee, 2022).

Notably, in our systematic review, we documented promising evidence of the benefits of SEL programs specifically for racially marginalized youth in U.S. schools. After participating in the *BASE* program, racial and ethnic minority youth demonstrated increased prosociality and reductions in defiant behavior, comparative to their white peers (Dawes et al., 2020). African American students who participated in *Making Choices* demonstrated reductions in overt and relational aggression at school relative to their white peers (Wike, 2011). And, in *It's Your Game... Keep it Real*, differential benefits for Hispanic and African American students were documented, including reductions in dating violence victimization and perpetration for Hispanic students and a decrease in physical dating violence victimization for African American students (Peskin et al., 2014), among other promising results in the corpus of studies reviewed. Although the scale of the evidence available is limited, there are promising pockets of evidence that point to both the differential and beneficial effects for students with marginalized racial and ethnic identities in the United States. Evolved models of measurement and identification that are sensitive to

and inclusive of all student experiences are necessary to advance research and practice benefiting marginalized students. Such models would move away from a deficit frame of marginalized students to increasingly articulating the role of racism, rather than race, in the conceptualization of and analysis of differential effects (Chokshi et al., 2022).

We write these words with an abundance of caution. The stakes of representation are high for students whose marginalized identities are also threatened within society. According to Title IX federal regulations and the U.S. Department of Education, policies and behaviors within American schools that discriminate based on student race, language, gender, or disability identity are prohibited (U.S. Dept of Education, 2021). The mandate's claim that schools must ensure that all students feel safe and supported within their school, and that they have equal access to educational opportunities. The juxtaposition of Title IX and its best intentions alongside the increasing hostile conditions for learning in the U.S. public schools at the time this paper was written is striking.

Consider the experience of a gender or sexually marginalized adolescent, such as an adolescent who is gender expansive and identifies as a member of the LGBTQIA+ community in the United States. This student's identity may not be safe to share due to discriminatory policies at the school or state level, by sheer function of their zip code, and/or they may not be out or be already experiencing rejection by their family. As mounting policies and hate speech become policy across the United States (Heim & Rozsa, 2022; Pollock et al., 2022; Ridings, 2020; Swidriski, 2022), we are not providing students the message that it is safe to be themselves at school (Pollock et al., 2022; Rosky, 2017; Swidriski, 2022). As a research community, we cannot expect students to opt in to be seen at a time when an increasing number of schools, communities, and their country may be actively working to omit them. Indeed, in our analyses, we were able to explore differential effects by gender in the binary (male/female), and from which determined that boys and girls benefited equally from SEL programs. The limitation of gender presentation only in the binary inhibits our ability to understand if and how SEL programs support or differentially affect gender expansive youth, failing to account for an estimated 10% of the U.S. student population and their experiences with SEL (Kidd et al., 2021).

Limitations

We situate our findings in the context of the limitations of this review. First, this review is only based on studies from 2008 through 2020 that had been previously identified to have been conducted in the U.S. and have met strict inclusion criteria for analyses. It is possible that inclusion of studies prior to 2008 (such as the 213 included in Durlak et al., 2011) would inform our findings. Future research will explore this question among others. In addition, the evidence reported for each specific SEL program in this review is not exhaustive; strict inclusion criteria limited the data presented here to only studies that allowed for the calculation of effect sizes on student outcomes. As such, findings should not be generalized to advance or disadvantage any one SEL program, nor suggest limitations in the evidence available for an SEL program, respectively. Rather, findings can be understood as a snapshot of the most rigorous experimental evidence available for U.S. SEL programs to date, and limitations in the data reported can serve as a call in to U.S. SEL program evaluations to be more explicit and thorough in their respective and collective reporting on the evidence for SEL. Lastly, given the nature of including randomized control trials and quasi-experimental designs, methodological precedence in the aggregation of effects has not historically encouraged the level of discrete representation we are seeking to analyze as a part of this study. Although it will take time for the systems of 'best practice' in experimental designs of educational research to catch up to heterogeneity of student identities in the U.S. classroom, there is growing evidence regarding variations in SEL programming, implementation, and outcomes for youth through qualitative investigations (see Elias & Yuan, 2020; Osher et al., 2016;

Rivas-Drake & Umana-Taylor, 2019; Rosario-Ramos et al., 2021). Wilson & Anagnostopoulos, 2021 methodological guidance paper on conducting a qualitative review is a helpful resource for the field to consider supporting the most complete representation in the evidence for USB SEL.

A call to action

There is robust evidence for the effectiveness of SEL for students K-12 in the United States. Systematic analysis of more than a decade of experimental studies of universal school-based SEL programs confirms the benefits of SEL on student academic, social, and emotional functioning at school. Programs taught by teachers produce the strongest effects for students, and the potential for students to benefit from SEL at scale in the United States is evidenced by a robust and ever-growing literature of experimental studies.

Notably, the U.S. student body continues to increase in heterogeneity at the same time SEL programs have differentially evolved. Intentional considerations for representing students with marginalized identities in the United States will support more robust and complete evidence of the effectiveness of SEL programs for all students, classrooms, and communities at the level of nuance and differentiation warranted to honor the unique and multidimensional experiences of students in U.S. schools. Once more, as the empirical evidence for SEL grows, we acknowledge that we prepared this manuscript at a time of mounting public criticism and pushback on adopting SEL in schools explicitly. This growing opposition to SEL has fueled action that stands to harm marginalized students – specifically racially marginalized, LGBTQ+, and female students. These are the same populations of students who are currently most susceptible to diminished mental health and suicidal ideation in the United States (CDC, 2023).

As researchers, SEL advocates, students, parents, and partners, we come to this work with an obligation to ensure that the possibility and promise of SEL is actualized for all students – and particularly inclusive of students with marginalized racial, ethnic, linguistic, gender, sexual, and ability identities. In light of the robust evidence of effectiveness available for SEL in the United States, SEL science will benefit from increased specificity, intentional representation, and open science data practices in the service of all youth.

Declaration of Competing Interest

We have no known conflicts of interest to disclose.

Acknowledgements

We wish to thank Drs. Michael Strambler, Lauren Naples, and Megan Kirk Chang, and colleagues Maegan Genovese, Victoria Mack, and Victoria McClare for their support.

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